

IVB Model Documentation

Initial Balance Breakout Strategy — Design, Logic, Results & Roadmap

0. Overview

The IVB (Initial Balance Volume) model is a directional intraday futures strategy built on top of the Opening Range Breakout concept, enriched with volume profile analysis and order flow confirmation. It was originally inspired by Fabio Valentini's IVB model.

The strategy has two layers:

- **Directional framework** — identical across all model versions: define the initial balance, detect a breakout, wait for a retest of the reload zone.
- **Entry confirmation** — this is where the models differ. Model 1 used a simple delta candle. Model 2 v2 requires an absorption candle followed by a momentum entry candle.

The current production strategy is `ivb_model2_v2.py`. It supports 30+ instruments via automatic tick size injection from `ASSET_INFO`, and includes direction flipping, a peak-based volume profile, and a notes column for trade inspection.

1. Core Concept

The first 30 minutes of the New York cash session (09:30–10:00) represent the most important price discovery window of the day. Institutional participants battle for directional control during this window. Whoever wins defines the session bias.

The IVB model tracks this battle and aligns with the winner:

- If buyers break out above the IB range first → look for long entries on the retest.
- If sellers break below the IB range first → look for short entries on the retest.

The volume profile of the initial balance provides the precise entry zone, stop loss level, and defines what constitutes invalidation of the trade idea.

The three-layer framework:

Layer	What it provides	How it's implemented
1. Direction	IB breakout tells you who won the most important battle of the day	First tick of the day above IVB_high or below IVB_low
2. Location	Volume profile defines where the reload zone is	AIWPC/Vol profile Base expected profile center
3. Confirmation	Order flow at the reload zone confirms the counter-trend	Absorption candle with wick + anomalous counter-trend volume at a single tick

2. Step-by-Step Logic

Every trading day goes through the following pipeline. Steps 1–4 are the directional framework. Steps 5–9 are the entry and trade management layer.

Step 1 — Define the Initial Balance

Scan the first `ib_minutes` bars of RTH (default: 30 minutes, 09:30–09:59 inclusive).

- `IVB_high` = highest high in that window.
- `IVB_low` = lowest low in that window.
- `IVB_range` = `IVB_high` - `IVB_low`. If zero, skip the day.

Note: `ib_minutes` = 30 means bars 09:30 through 09:59 are included. The 10:00 bar is NOT part of the IB — it is the first post-IB bar. `ib_minutes` can be set to 15 (scalp bias) or 60 (intraday bias).

Step 2 — Build the Volume Profile

Compute a fixed-range volume profile using `tick_volume` data from the IB bars only. This uses the peak-based algorithm (see Section 4 for full detail).

- **POC** — price level with the highest raw volume inside the value area.
- **VAH** — value area high (top of the 70% volume zone).
- **VAL** — value area low (bottom of the 70% volume zone).

The IB volume profile is computed once per day and is reused for all flips. It never changes during the session.

Step 3 — Detect the Breakout

After the IB closes, scan post-IB bars for the first breakout of the IB range.

- **Long breakout:** first bar that CLOSES above `IVB_high`.
- **Short breakout:** first bar that CLOSES below `IVB_low`.
- Whichever happens first defines the direction for the day.
- If no breakout occurs → no trade.

Close required: A wick that touches the level but closes back inside is NOT a breakout. This filters out false breakouts driven by a single aggressive order.

Step 4 — Wait for the Retest

After the breakout, wait for price to pull back toward the IB value area — the reload zone.

- **Long:** bar low touches or enters the VAH zone (`bar low <= VAH`).
- **Short:** bar high touches or enters the VAL zone (`bar high >= VAL`).
- The retest must occur within `retest_window` bars (default: 30) after the breakout.
- If no retest within the window → no trade.

Step 5 — Find an Absorption Candle

After the retest is detected, scan the next entry_window bars (default: 25) for an absorption candle. An absorption candle is evidence that the counter-move is being rejected at the reload zone.

Absorption candle definition (long bias):

1. Lower wick $\geq$ wick_threshold fraction of total bar range (default 0.3 = 30%):

```
lower_wick = min(open, close) - low
wick_ratio = lower_wick / (high - low)
wick_ok    = wick_ratio  $\geq$  wick_threshold
```

2. At least one price level inside the wick zone has sell volume $\geq$ absorption_mult x rolling baseline (default: 2.0x):

```
# For each price level between low and min(open, close):
sell_vol_at_level  $\geq$  rolling_avg_sell_per_tick * absorption_mult
```

3. Candle body direction is **irrelevant** — a bullish candle with a large lower wick and absorbed sellers is equally valid as a bearish one.

Short bias mirror:

- Upper wick $\geq$ wick_threshold of bar range.
- Buy volume at a single level in the upper wick $\geq$ absorption_mult x rolling baseline.

Rolling baseline:

Computed on RTH bars only, starting at 09:35 (first 5 minutes excluded). Per bar: sell_volume / range_ticks. Rolling mean of last absorption_window bars (default: 20). Requires a full window before grading any candle — avoids noisy early-session baselines.

Absorption level:

Once an absorption candle is found, its low (long) or high (short) is recorded as the absorption level. This level is used for invalidation tracking.

Step 6 — Find the Entry Candle

After absorption is confirmed, scan up to entry_after_absorption bars (default: 5) for a momentum entry candle. The entry candle must satisfy:

- **Body threshold:** $|close - open| / (high - low) \geq body_threshold$ (default: 0.5). The body must cover at least 50% of the bar range.
- **Delta threshold:** volume_delta_pct $\geq$ delta_threshold for longs, $\leq -delta_threshold$ for shorts (default: 30.0%). Strong directional order flow required.

Absorption invalidation during entry scan:

If any bar in the entry scan closes below the absorption level (long) or above it (short), that absorption candle is invalidated. The strategy discards it and resumes scanning for a new absorption candle from the current position in the entry window.

Whole trade invalidation:

If any bar closes below VAL (long) or above VAH (short) at any point during the entry window, the entire trade idea is invalidated for this direction. This triggers either a direction flip (if flips remain) or a no-trade for the day.

Entry execution:

Entry = next bar's open after the valid entry candle. You cannot enter at the close of the signal bar in live trading — the strategy respects this by using the following bar's open.

Step 7 — Stop Loss

Controlled by `sl_type` parameter:

sl_type	Rule	Notes
0	Stop at VAL (long) or VAH (short)	Fixed structural level. If price closes through the full value area the thesis is wrong.
1	Swing low/high of all retest bars up to entry but noise-sensitive. Testing showed 79% stop rate on ES/NQ — not recommended.	

Recommended: `sl_type = 1` (swing low) works well with the absorption entry because you are entering much closer to the actual turning point. `sl_type = 0` (VAL) was the correct choice for Model 1 but absorption entries produce tighter swing lows.

Step 8 — Take Profit

```
risk = abs(entry_price - sl)
tp   = entry_price + (rr * risk) # long
tp   = entry_price - (rr * risk) # short
```

Fixed risk:reward ratio controlled by `rr` parameter (default: 1.0, best results at 2.0). Increasing RR beyond 2.0 reduces win rate faster than it increases avg win — the market does not extend far enough often enough to justify higher targets.

Step 9 — Trade Execution and Exit

The trade runs until one of the following exits fires:

Exit reason	Condition
tp	High (long) or low (short) touches the TP level within timeout window
sl	Low (long) or high (short) touches the SL level within timeout window
eod	Session ends (16:00 NY) without hitting either level
tp_timeout	trade_timeout bars elapsed and trade is in profit — exit at current close
sl_timeout	trade_timeout bars elapsed, trade in loss, adjusted SL is hit

Timeout logic:

If `trade_timeout` bars elapse without hitting SL or TP:

- In profit → exit immediately at current close (`exit_reason = tp_timeout`).
- In loss → move TP to entry (break even), tighten SL to lowest low since entry. Let trade run to EOD.

Best results: `trade_timeout = 999` (effectively disabled). The timeout logic was hurting performance by cutting winning trades short. Set to 120 in current best-result params as a safety net.

Step 10 — Direction Flipping

When a trade idea is invalidated (price closes through VAL for longs, VAH for shorts), instead of giving up on the day, the strategy attempts to flip direction and find a trade in the opposite direction using the exact same ruleset.

Flip logic:

1. Flip direction (long → short, or short → long).

2. Resume scanning from the invalidation bar onward.
3. Run `_detect_breakout()` in the new direction from this point.
4. If a breakout is found in the correct direction → run retest, absorption, entry as normal.
5. If the first breakout found is the wrong direction → no trade.
6. Repeat up to `max_flips` times (default: 4).

IB profile reuse:

The same POC/VAH/VAL computed from the IB bars is reused for all flips. The value area doesn't change — only the direction and the scan start position change.

Why this works: A close through VAL means the entire value area has been rejected from above — sellers are in control. This is itself a short signal. Flipping captures this without requiring a completely new setup.

3. Parameters

Parameter	Default	Best Result	Description
ib_minutes	30	30	IB range duration in minutes (15, 30, or 60)
delta_threshold	30.0	30.0	Min volume_delta_pct for entry candle
body_threshold	0.5	0.5	Min body/range ratio for entry candle
rr	1.0	2.0	Fixed risk:reward ratio
sl_type	0	1	0 = VAL/VAH, 1 = swing low/high
retest_window	30	40	Max bars to wait for retest after breakout
entry_window	15	25	Max bars to scan for absorption after retest
entry_after_absorption	5	5	Max bars to scan for entry candle after absorption
trade_timeout	999	120	Bars before timeout logic triggers (999 = disabled)
max_flips	4	4	Max direction flips per day after invalidation
wick_threshold	0.4	0.3	Min wick/range ratio for absorption candle
absorption_mult	2.0	2.0	Wick level volume must be >= this x rolling avg
absorption_window	20	20	Rolling N bars for sell_per_tick baseline
tick_size	0.25	—	Auto-injected from ASSET_INFO. Never set manually.

tick_size is in HIDDEN_PARAMS — it is never shown as a UI widget. The backtester automatically injects the correct value from ASSET_INFO based on the selected asset.

4. Volume Profile Algorithm

The IVB strategy uses a peak-based volume profile algorithm. This replaces the old simple max-volume POC approach which failed on bimodal profiles by producing a value area spanning the valley between two peaks.

- **Step 1 — Build raw levels:** Aggregate tick_volume JSON across all IB bars into {price: total_volume}.
- **Step 2 — Smooth:** Apply a 3-tick rolling average to volumes (window of ± 1 tick). Removes single-tick noise spikes without changing the shape of the distribution.
- **Step 3 — Find local maxima:** A peak is where $\text{smoothed}[i] > \text{smoothed}[i-1]$ AND $\text{smoothed}[i] > \text{smoothed}[i+1]$. Equivalent to finding where the discrete derivative flips from positive to negative. Endpoints checked separately. Flat profiles fall back to the global max.
- **Step 4 — Cluster peaks:** Peaks within 4 index positions (4 ticks) of each other are merged into one cluster. The highest smoothed value per cluster survives. Top 5 clusters taken.
- **Step 5 — Refine to raw peak:** For each cluster peak, scan ± 3 ticks in the raw (unsmoothed) volumes. The index with the highest raw volume becomes the POC candidate. Ensures the final POC reflects actual traded volume, not the smoothed estimate.
- **Step 6 — Expand value area:** For each POC candidate, expand outward — always adding the side with higher adjacent volume — until cumulative volume $\geq 70\%$ of total. Record VAH and VAL.
- **Step 7 — Pick tightest:** Compare VAH - VAL across all candidates. The candidate with the smallest price range wins. On bimodal days this selects the densest cluster, not the global max.

- **Step 8 — Refine POC within VA:** After selecting the winning VA, scan all price levels between VAL and VAH in the raw profile. The level with the highest raw volume becomes the final POC. This corrects cases where the smoothed peak index and the true highest-volume tick differ.

5. Output Schema

The strategy returns a DataFrame with these columns. The backtester computes ticks from pnl_points — the strategy never stores ticks.

Column	Type	Description
date	date	Trade date
direction	str	lowercase 'long' or 'short'
entry_time	Timestamp	Bar open time of entry (next bar after signal candle)
exit_time	Timestamp	Bar time of exit
entry_price	float	Open of the entry bar
exit_price	float	SL, TP, or close price at exit
sl	float	Stop loss price level
tp	float	Take profit price level
exit_reason	str	tp / sl / eod / tp_timeout / sl_timeout
pnl_points	float	exit_price - entry_price (long), entry_price - exit_price (short)
notes	JSON str	Strategy metadata (see Section 6)

6. Notes Column

Every trade record includes a JSON string in the notes column. The backtester parses this and renders it as a second metrics row in the trade detail view.

```
{
  "breakout_time": "2024-03-15 10:03:00-04:00", # timestamp of breakout bar
  "retest_time": "2024-03-15 10:11:00-04:00", # timestamp of first retest bar
  "absorption_time": "2024-03-15 10:14:00-04:00", # timestamp of absorption candle
  "flip_count": 0, # number of direction flips before entry
  "ivb_high": 5102.25, # IB range high
  "ivb_low": 5094.50, # IB range low
  "poc": 5097.75, # point of control
  "vah": 5100.25, # value area high
  "val": 5095.50 # value area low
}
```

In the backtester UI, the notes row shows: Breakout time / Retest time / Absorption time / Flip count / POC / VAH / VAL — all in the same 7-column row as the main trade stats.

7. Performance (ES, 1 Year)

Backtested on ES enriched 1-minute candles (tick_volume + passive_orders required). 1 year of data. Best parameter set shown.

Metric	Model 1 (delta only)	Model 2 v2 (absorption)
Total trades	~80	68
Win rate	~51.5%	54.4%
Avg win	~65 ticks	62.3 ticks
Avg loss	~49 ticks	-30.0 ticks
Profit factor	~1.0	2.65
Sharpe	~0.5	—
Avg win / avg loss	~1.33x	2.08x

The absorption filter cut ~55% of trades from Model 1. The remaining trades have meaningfully better avg win/loss ratio (2.08x vs 1.33x) and win rate (54.4% vs 51.5%), confirming the filter is selecting higher-quality setups rather than just reducing sample size.

Monte Carlo result (best params, \$100k account):

Metric	Value
Median final equity	\$145,531 (+45.5%)
Mean final equity	\$146,041 (+46.0%)
Final equity 1s (68%)	\$128,719 – \$163,320 (+28.7% to +63.3%)
Final equity 2s (95%)	\$114,798 – \$180,040 (+14.8% to +80.0%)
Final equity 3s (99.7%)	\$103,223 – \$198,594 (+3.2% to +98.6%)

Limitation: Only 1 year of data available (ES/NQ enriched tick_volume dataset). 68 trades is a real but thin sample. Walk-forward validation requires multi-year data. Results should be treated as promising but not statistically conclusive.

8. Data Requirements

The IVB strategy requires enriched 1-minute candles with tick_volume and passive_orders columns. Standard OHLCV data is NOT sufficient.

Column	Required	Used for
open/high/low/close	Yes (float64)	IB range, breakout detection, bar structure
volume	Yes	Volume profile construction
buy_volume	Yes	Rolling baseline (buy_per_tick)
sell_volume	Yes	Rolling baseline (sell_per_tick), absorption grading
volume_delta	No	Not used directly (delta_pct is used)
volume_delta_pct	Yes	Entry candle delta threshold check
tick_volume	Yes	Volume profile algorithm + absorption wick density check
passive_orders	No	Not used in current model — reserved for future passive entry type

Non-equity assets: tick_volume and passive_orders enriched data is only available for ES/NQ from the Databento subscription period. Other assets have OHLCV-only data. The IVB strategy will not run on OHLCV-only datasets.

9. Known Issues and Limitations

- **Session times hardcoded:** RTH is hardcoded as 09:30–16:00 NY. This is wrong for CL (pit session 09:00–14:30), rates, FX, and metals. Results on non-equity assets are invalid until per-asset session times are implemented.
- **Thin sample size:** 68 trades over 1 year. Not enough for statistically robust optimization. Walk-forward validation is blocked until multi-year enriched data is available.
- **Rolling baseline warmup:** absorption_window = 20 bars, starting at 09:35. The baseline is not valid until approximately 09:55. Absorption candles before this window cannot be graded — they are skipped.
- **Single absorption type:** Current model only detects wick-based absorption. Absorption where sellers hit the bid in the final seconds of a bar (no visible wick) is not detected.
- **Passive orders unused:** passive_orders column is loaded but not used in the current entry logic. Reserved for the future passive order entry type.
- **Flip direction mismatch:** After a flip, if _detect_breakout() returns a breakout in the wrong direction, the strategy returns no trade. This may be too conservative — a breakout in the original direction could be re-evaluated.
- **No multi-timeframe context:** The strategy operates entirely on 1-minute bars. Higher timeframe context (daily range, prior session high/low, weekly levels) is not incorporated.

10. Development Roadmap

Completed

- ✓ pnl_points only in output — backtester converts to ticks via ASSET_INFO
- ✓ Peak-based volume profile algorithm — replaces simple max-volume POC
- ✓ POC refinement within VA — highest raw volume tick inside the value area
- ✓ Absorption candle grading — wick threshold + single-level volume density
- ✓ Direction flipping — up to max_flips per day when trade idea invalidated
- ✓ Notes column — breakout_time, retest_time, absorption_time, flip_count, IB levels
- ✓ HIDDEN_PARAMS — tick_size auto-injected from ASSET_INFO, not shown in UI
- ✓ Chart view settings in backtester — candles before entry or fixed session time

Planned

More entry types

- Consecutive absorption candles — 2+ absorption candles in a row as a stronger signal.
- Passive order entry — large resting bid/ask at a single tick level as a standalone entry signal, with a separate (lower) multiplier threshold since passive and aggressive volume scale differently.
- Exhaustion entry — counter-move progressively weakens (shrinking delta, volume, bar size) before a reversal candle. Details to be determined from chart observation.

Earlier entries

- Entering on or immediately after the absorption candle without waiting for a full delta/body confirmation candle. Current lag sometimes means entering at the end of an expansion move. Approach TBD — needs chart-level analysis of losing trades.

Minimum risk threshold — higher TP

- When the SL is too close to entry (risk in ticks below a minimum threshold), instead of taking a low-quality trade or skipping it, extend the TP to a higher RR. Prevents entering trades where normal bar volatility will stop out before the move develops.

Trail SL to absorption level

- Once the trade is in profit by X ticks, move the SL up to the absorption candle low (long) or high (short). Locks in some profit while giving the trade room to run. Trigger mechanism TBD.

Better TP targeting

- Current TP is a fixed RR multiple of risk. Future ideas include: prior session high/low as target, IB range extensions (1x, 1.5x, 2x above IVB_high), next significant volume profile level. Approach TBD.

Session times per asset

- RTH hardcoded to 09:30–16:00 NY must be replaced with per-asset session config. Required before the strategy can be meaningfully tested on CL, GC, ZN, FX assets.

Walk-forward validation

→ Requires multi-year enriched tick_volume data. Currently blocked by data availability. Split in-sample / out-of-sample to verify params generalize.

Note: All future improvements are documented here as ideas, not specifications. The exact rules for each will be determined through chart observation and backtesting rather than being designed in advance.